

PDF-Based RAG System: Technology Stack

Tools and their purpose

Part	Tool / Technology	Purpose
Programming	Python	Main backend/AI development
PDF Processing	PyMuPDF / fitz	Extract text from PDF
NLP	NLP libraries	Clean and process extracted text
Text Chunking	LangChain or custom Python	Split PDF into meaningful chunks
Embeddings	Sentence Transformers	Convert chunks into numerical vectors
Vector Database	FAISS or Chroma	Store and search embeddings
RAG	LangChain / LlamaIndex or custom pipeline	Retrieve relevant PDF content before generation
LLM	GPT / Gemini / Llama / other LLM	Generate answers, quizzes, revision material
Frontend	Streamlit / React	Build the user interface
Backend / API	FastAPI / Flask	Connect frontend with AI pipeline
Evaluation	SQuAD 2.0	Evaluate QA/RAG performance
Source Tracking	Metadata	Store page number/chunk information for references